

Midterm - 2/15/2017

- Points pre-assigned to review questions and problems are just an indication of the final score: partial and unsatisfactory answer/solution will receive a fraction of the points, whereas excellent answers may exceed the indicated points.
- Use of books/notes is forbidden.
- Indicate name and UC ID number in ALL the solution sheets. Number the solution sheets and indicate total number of sheets (*e.g.*, first sheet 1/3, second sheet 2/3, third sheet 3/3).

R1 (5 Points) Consider two random variables X_1 and X_2 , distributed according to an exponential distribution with parameters λ_1 and λ_2 . The expectation of the random variable $X = \min(X_1, X_2)$ is

- a) $\lambda_1 + \lambda_2$;
- b) $\min(\lambda_1, \lambda_2)$;
- c) $1/(\lambda_1 + \lambda_2)$**
- d) $1/\min(\lambda_1, \lambda_2)$;

R2 (10 Points) A Poisson process generates events at rate λ . Each event is discarded with probability p . What is the rate of the resulting process?

- a) $p\lambda$
- b) p/λ
- c) $(1 - p)\lambda$**
- d) $(1 - p)/\lambda$

R3 (5 Points) List and describe the sources of delay in the Internet.

Answer

Processing delay (1 pt): The time to examine the packet's header and determine where to direct the packet. There could be other types of processing, *e.g.* checking for bit-level errors.

Queuing delay (1 pt): The time for the packet to wait for earlier-arrived packets to be transmitted. It depends on the packets in the queue when the packet arrives.

Transmission delay (1 pt): The time to transmit the packet onto the link. It depends on the length (size) of the packet and the transmission rate of the link.

Propagation delay (1 pt): The time for the packet to travel (propagate) through the link physically. It depends on the distance and the propagation speed on the link.

R4 (10 Points) Briefly describe the salient differences between packet switching and circuit switching. Why is the circuit switching model used in the design of the telephone network?

Answer

Difference (4 pt): In circuit switching, a physical link is established and dedicated for the call (session). All traffic goes through that link. In packet switching, data are grouped into small chunks (packets). Each packet is switched and routed separately.

Circuit switching is used for telephone network, because **(1) (3 pt)** phone calls are sparse, but generate data at relatively constant rate during the session, **(2) (3 pt)** circuit switching provides stable quality of service, given the predictable rate.

R5 (5 Points) What is the main source of packet loss in packet switched wired networks?

Answer

Packets are **dropped (1 pt)** at **full (1 pt)** buffers (queues).

R6 (10 Points) Concisely describe the client-server and the peer-to-peer application architectures and highlight their differences.

Answer

The client-server architecture requires **always-on infrastructure servers (2 pt)** that all users (clients) connect to. In a P2P architecture, there is minimal (or no) reliance on always-on infrastructure servers. Instead, pairs of intermittently connected hosts, called peers, **communicate directly with each other (2 pt)**.

R7 (10Points) Describe the non persistent and persistent connection approach in HTTP. What are the pros and cons of the two approaches?

Answer

Description (4 pt): In **non-persistent HTTP**, each request/response pair is sent over a *separate TCP connection (1 pt)*. In **persistent HTTP**, all of the requests and their corresponding responses are sent over the *same TCP connection*.

Non-persistent HTTP allows transmission of multiple objects at the same time, but requires TCP connection setup for each object, which results in bigger overhead. Persistent HTTP transmits the objects back-to-back in one TCP connection, which has less overhead, but the connection is kept alive until it times out.

R8 (15 Points) List and describe the layers of the Internet stack.

Answer

Four-layer Internet stack (8 pt): Link layer, network (Internet or IP) layer, transport layer, and application layer (2 pt each). The link layer of the Internet stack covers the physical layer and the data link layer in the OSI model.

Refer to **Kurose-Ross** Section 1.5 for description of each layer.

R9 (15 Points) Briefly describe the performance requirements of real-time multimedia applications. Why is Forward Error Correction preferable to packet retransmission?

Answer

Performance requirements (8 pt): High bit rate (bandwidth), small jitter, but tolerant to non-significant delay. Content can be compressed for quality-bandwidth trade-off.

FEC increases the effective systems throughput, even with the extra check bits added to the data bits, by eliminating the need to retransmit data corrupted by random noise, which is non predictable and causes jitter.

R10 (15 Points) Describe time division multiplexing in digital telephone networks.

Answer

Time division multiplexing (TDM) is a communications process that transmits two or more streaming digital signals over a **common channel (5 pt)**. In TDM, incoming signals are divided into equal fixed-length **time slots (5 pt)**. After multiplexing, these signals are transmitted over a shared medium and reassembled into their original format after de-multiplexing.